

Unit 4: Data Collections

Introduction to Using Data Sets

Aligned to:

AP Computer Science A Course and Exam Description
Effective Fall 2025

Topic 4.2

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What Is a Data Set?

- A data set is a collection of specific pieces of information.
- Test scores for a class. Daily rainfall for a year. Every transaction a shop made last month.
- What makes it a data set rather than a pile of numbers is that the pieces are related and can be examined together to answer a question.

How We Work With One

- You almost never look at a data set all at once. You work through it.
- Values in the set are accessed and used ONE AT A TIME, and each one is processed toward whatever outcome you are after.
- That pattern — visit each value, do something with it, carry a result forward — is a traversal, and it is what the rest of this unit is about.
- Arrays, ArrayLists and files are three ways to hold a data set. The traversal barely changes.

Plan Before You Code

- Data can be represented in a chart or a table. That picture is not decoration — it is how you plan the algorithm.
- Laying the data out shows you what one row is, which column you care about, and what you are accumulating as you go.
- Describe the algorithm in plain language first. If you cannot say it in English, the Java will not save you.

A Data Set to Work With

- A cafeteria waste audit, five school days:

Day	Meals served	Pounds discarded
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1	412	38
2	398	51
3	405	33
4	421	47
5	388	62

- Each row is one day. Each column is one thing measured about that day.

From Table to Algorithm

- Question: which day wasted the most food per meal served?
- In words, before any Java:
 - 1) Start with no best day yet.
 - 2) For each row, compute pounds discarded divided by meals served.
 - 3) If that is bigger than the best so far, remember this day instead.
 - 4) When the rows run out, the remembered day is the answer.
- That description is the algorithm. Writing it in Java is now translation, not invention.

Three Questions

- When you are staring at a data set and do not know where to start, answer these:
- 1) What am I looking at each step? (one row? one number?)
- 2) What am I carrying with me? (a running total? a count? the best so far?)
- 3) What do I do when the data runs out? (return it? divide it? print it?)
- Almost every algorithm in this unit is those three answers in a loop.

Algorithms to Know

- Over a data set, you will be asked to:
- total the values
- count the ones that meet a condition
- find the largest or smallest
- compute an average
- find the item with the biggest something (not the biggest value itself)
- check whether every value, or any value, satisfies a test
- Learn the shape of these once and the specific question stops mattering.

Your Turn

- Using the cafeteria table, describe each of these in words — carrying, comparing, and finishing:
- 1) What was the average number of meals served per day?
- 2) How many days had more than 45 pounds of waste?
- 3) Which day had the best ratio of meals served to pounds discarded?
- Do not write Java yet. Write sentences. Then compare with a partner — where you disagree is where the specification was ambiguous.

References

- 1) Building Java Programs: A Back to Basics Approach by Stuart Reges and Marty Stepp
- 2) Runestone CSAwesome Curriculum:
<https://runestone.academy/runestone/books/published/csawesome/index.html>

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<https://longbaonguyen.github.io>